

Workflow and architecture patterns for working together 2: Architectures

Jacob Archambault

May 29, 2026

Outline

1 Current challenges and their canonical solutions

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- 2 Branching: theory and statistics
 - Theory
 - Statistics
 - Devops Statistics recapitulation
 - Approximating DevOps metrics in our current setup
 - Velocity
 - Safety
 - Maintenance to value-adding work percentage

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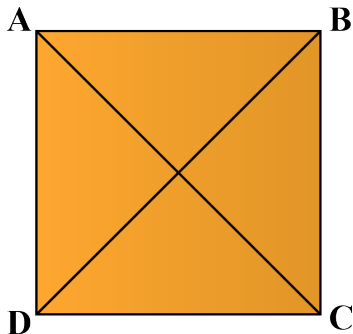
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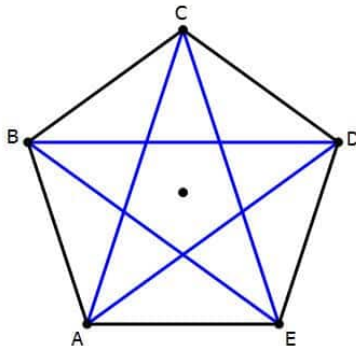
1975: The Mythical Man Month, Fred Brooks

- number of direct communication paths for n individuals= $n(n-1)/2$



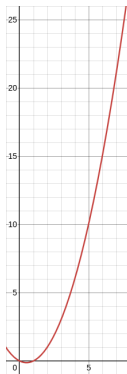
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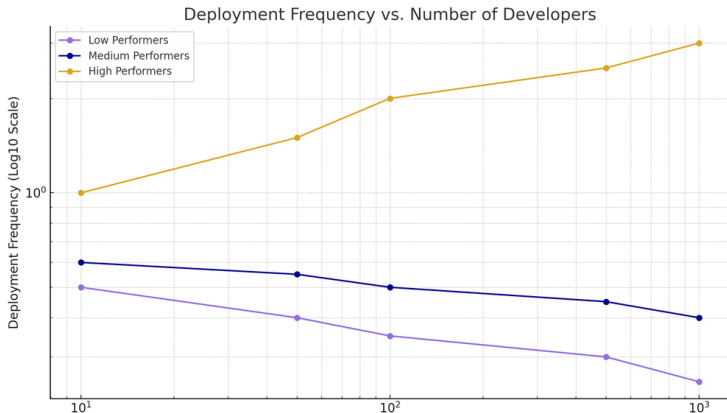
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$$n(n-1)/2$$



The Mythical Man Month, Fred Brooks (cont.)

- corollary: adding more people to a project can lead not only to diminishing returns on delivery speed, but to objectively less work being completed



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A naive probability theory for merge consistency: modeling branches as sets of commits

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- $= \frac{1}{25}$

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A naive probability theory for merge consistency: general definition for distinct branches diverging from a common ancestor

- Let A be the set of unmerged commits across k distinct branches $A_1 \cdots A_k$, where $A = A_1 \sqcup A_2 \sqcup \cdots \sqcup A_k$, $n = |A|$, and $n_i = |A_i|$ for $1 \leq i \leq k$. Then

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$$P(\text{no conflicts}) = \frac{1}{2^n - (\sum_{i=1}^k 2^{n_i}) + k}$$

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The State of DevOps Report: throughput metrics

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The State of DevOps Report: stability metrics

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The State of DevOps Report: results

'Astonishingly, these results demonstrate that there is no trade-off between improving performance and achieving higher levels of stability and quality. Rather, high performers do better at all of these measures. This is precisely what the Agile and Lean movements predict, but much dogma in our industry still rests on the false assumption that moving faster means trading off against other performance goals, rather than enabling and reinforcing them.'
- Nicole Forsgren, Jez Humble, and Gene Kim, Accelerate (2017), p. 14

The State of DevOps Report: results

When compared to low performers, elite performers realize

127x

faster lead
time

182x

more
deployments
per year

8x

lower change
failure rate

2293x

faster failed
deployment
recovery times

- 2024 State of DevOps Report

Approximating DevOps metrics in our current environment

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Throughput: overall velocity metrics

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NG_Claims_837D_COB	74	23 Mar	5
PEDI-35120_837D_COB	85	15 Apr	5
Either COB branch	159	23 Mar	6
All other branches	123	23 Mar	12
	72	15 Apr	10
	266	21 Jan	

Source: GH Actions Simulation tests, 28 May 2026

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Stability: pipeline failure rate metrics

Branch	Simulation test failures	Unit test failures	Earliest commit
NG_Claims_837D_COB	56/74 (76%)	56/74 (76%)	23 Mar
PEDI-35120_837D_COB	20/85 (23.5%)	21/85 (24.7%)	15 Apr
Either COB branch	76/159 (48%)	77/159 (48%)	23 Mar
All extant branches	73/123 (59%)	87/123 (71%)	23 Mar
	50/72 (69%)	51/72 (71%)	15 Apr
	92/161 (57%)	104/161 (64%)	21 Jan

Source: GH Actions, 28 May 2026

- 28% better simulation test pass rate since 23 Mar
- 76% better unit test pass rate since 23 Mar
- 150% better simulation test pass rate since 15 Apr
- 158% better unit test pass rate since 15 Apr

Stability metrics: merge work ratio

Branch	Merge commit % since 15 Apr
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All other branches	23/88 (26%)
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- Source command:

```
git log --since="2026-04-15"  
--oneline <--merges> <--all>  
<^PEDI-35120_837D_COB> | <grep -iv 'merge pull  
request'> | wc -l
```

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- Every change is tested by the test suite locally before committing

- tests are written against new code as that code is being written

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- daily PRs - stacked PRs

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- pipeline is definitive for deployment - daily stale branch job

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